

Analyzing Gaze Behavior During Emotion Recognition Using I-DT Fixation Detection

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Abstract

This project investigates how observers allocate visual attention when recognizing emotional facial expressions. Five participants viewed images of three individuals displaying five emotions (happy, sad, fear, neutral, angry) while their gaze was recorded using a Tobii 4C/5 eye tracker. We applied the Dispersion-Threshold Identification (I-DT) algorithm with two parameter settings to detect fixations, extracted fixation-based features such as mean fixation duration (MFD), and visualized gaze behavior using scanpaths, fixation overlays, and heatmaps. The analysis reveals systematic differences in fixation behavior across emotions and demonstrates how changing I-DT parameters affects the number and duration of detected fixations. Our results highlight both methodological considerations for fixation detection and substantive patterns in emotion-related gaze behavior.

CCS Concepts

• **Human-centered computing** → **Eye tracking**; • **Applied computing** → *Psychology*.

Keywords

eye tracking, fixation detection, I-DT, emotion recognition, mean fixation duration

1 Introduction

Gaze behavior provides a rich window into visual attention and cognitive processing, especially in social tasks such as interpreting facial expressions. Fixations—relatively stable periods when the eyes remain within a small region—are commonly used as indicators of information uptake. Their locations and durations can vary as a function of stimulus content, task demands, and individual differences [1, 2].

In this project we designed and implemented a controlled eye-tracking experiment to study how people visually explore emotional faces. Five observers viewed images of three individuals displaying five emotions: happy, sad, fear, neutral, and angry. We collected

gaze data using a Tobii 4C/5 eye tracker and focused on three main goals:

- (1) to characterize fixation patterns for each emotional condition;
- (2) to quantify differences in mean fixation duration (MFD) across emotions and observers;
- (3) to evaluate how the choice of parameters in the I-DT fixation detection algorithm influences detected fixations.

Our analysis is guided by the following hypothesis:

H1 (Feature-focused gaze). During emotion recognition, participants will concentrate their gaze on the eye and mouth regions of the face. Consequently, these regions will receive (a) a higher number of fixations and (b) longer mean fixation durations than other facial regions such as the forehead and cheeks.

The rest of this paper describes the experiment, the analysis pipeline, and the results. We follow the structure of a short scientific report: methods, algorithm description, data analysis, plots, and discussion, and conclude with a summary of group contributions.

2 Method

2.1 Participants

Five observers took part in the experiment: Kashif, Kidu, Ryan, Khang, and Zeeshan. All reported normal or corrected-to-normal vision and basic familiarity with computer displays. No personal identifying information was recorded. Participation was voluntary and uncompensated.

2.2 Stimuli

Stimuli consisted of static images of three different individuals (Person 1–3), downloaded from a publicly available facial expression dataset from Kaggle [3]. Each individual was photographed displaying five emotional facial expressions: happy, sad, fear, neutral, and angry. For each observer this resulted in $3 \times 5 = 15$ unique images, shown in Figure 1.

Images were displayed full-screen on a 1920×1080 pixel monitor with a mid-gray background. Faces were centered on the screen and scaled to maintain their original aspect ratio, with the eyes positioned approximately in the upper-central region of the display. Each stimulus image was presented for 12 seconds, followed by a 1-second inter-stimulus interval. A 2-second numerical countdown (2, 1) was displayed prior to each stimulus presentation to provide temporal preparation cues.

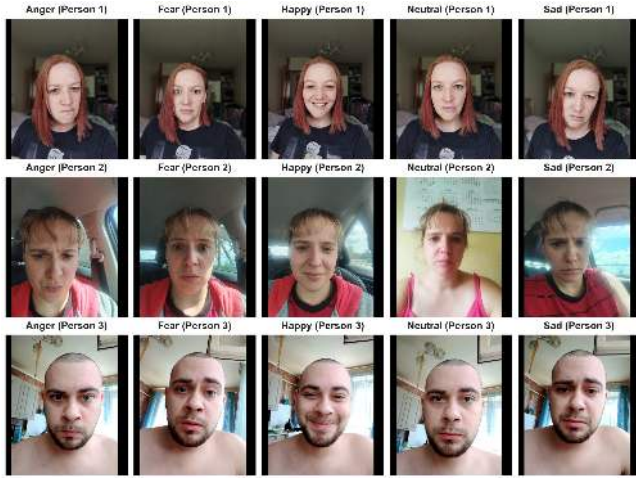


Figure 1: Dataset used: Three different individuals (Person 1–3), each photographed displaying five emotional facial expressions: Anger, Fear, Happy, Neutral, and Sad.

2.3 Apparatus and Procedure

Gaze data were recorded using the Hengam eye tracking system with a sampling rate exceeding 60 Hz. The eye tracker was mounted below a 1920×1080 pixel display (60 Hz refresh rate) with a mid-gray background. Following informed consent and eye tracker initialization, each trial began with a 2-second numerical countdown displayed centrally on the screen. This was followed by 12-second stimulus presentation, during which observers were instructed to “look naturally at the face and try to recognize the emotion displayed.” Gaze data were recorded continuously throughout stimulus presentation. Immediately after stimulus offset, participants provided a behavioral emotion classification response using a 5-category forced-choice format (Anger, Fear, Happy, Neutral, Sad) via keyboard input. A 1-second inter-stimulus interval separated consecutive trials. The complete experimental session, consisting of 15 trials (3 identities $\times$ 5 emotions), required approximately 4 minutes.

The data logged per sample included timestamp (microsecond precision), gaze position (x, y) in screen pixels, and a validity flag indicating data quality. Stimulus metadata (emotion category and image identifier) and behavioral responses were synchronized with gaze samples. We used static images and fixed trial durations to ensure precise synchronization between gaze data and stimuli, facilitating offline analysis of eye movement patterns during emotion recognition.

Source Code: Experiment design scripts

3 Fixation Detection Algorithm

3.1 Preprocessing

Raw gaze streams were cleaned before fixation detection. Samples with invalid tracking (validity value not equal to 1) were removed, and remaining points with coordinates outside the screen boundaries were discarded. Gaze samples were then grouped by observer and stimulus, and timestamps were verified to be strictly increasing within each trial to ensure temporal consistency.

3.2 Dispersion-Threshold Identification (I-DT)

To identify fixations we used the classic Dispersion-Threshold Identification (I-DT) algorithm [4]. The key idea is that a sequence of gaze samples belongs to a fixation if their spatial dispersion remains within a small region for at least a minimum duration.

Our implementation uses a sliding temporal window: starting from the first valid sample, we iteratively expand the window as long as the dispersion stays below the threshold. When the dispersion exceeds the limit, the window duration is evaluated; if it meets the temporal requirement, a fixation event is recorded.

3.2.1 Pseudocode.

Algorithm 1 I-DT Fixation Detection

```

1: Input: ordered gaze samples  $(x_i, y_i, t_i)$ 
2: Parameters: dispersion threshold  $D$ , minimum duration  $T_{\min}$ 
3: Initialize empty list of fixations
4:  $w \leftarrow$  first window of samples
5: while not at end of samples do
6:   compute dispersion of  $w$ 
7:   if dispersion  $\leq D$  then
8:     expand  $w$  by adding next sample
9:   else
10:    compute duration of  $w$ 
11:    if duration  $\geq T_{\min}$  then
12:      record fixation with centroid position and duration
13:    end if
14:    reset  $w$  starting at current sample
15:  end if
16: end while
17: process final window in the same way
    
```

All data cleaning, fixation detection, feature computation, and plotting were implemented in Python. The complete analysis scripts and notebooks are available to course staff at:

Source Code: Data Processing and Analysis Scripts

3.3 Parameter Settings

To study the effect of parameter choice we used two distinct I-DT settings:

- **Setting A (baseline):** dispersion threshold 50 pixels, minimum duration 100 ms;
- **Setting B (stricter):** dispersion threshold 30 pixels, minimum duration 150 ms.

Setting B imposes both stricter spatial and temporal requirements, so we expect it to produce fewer but more conservative fixations. Across all observers and stimuli, Setting A produced **1641**

fixations whereas Setting B produced **1477 fixations**, a reduction of roughly **10%**.

4 Data Analysis

Our analysis focused on fixation-based features and their variation across emotions, observers, and algorithm settings.

4.1 Mean Fixation Duration and Related Measures

For each observer and emotion condition we computed Mean Fixation Duration (MFD):

$$\text{MFD}_{s,c} = \frac{1}{N_{s,c}} \sum_{i=1}^{N_{s,c}} d_i,$$

where d_i is the duration of fixation i belonging to subject s and condition c , and $N_{s,c}$ is the number of such fixations. In addition, we computed:

- the standard deviation (SD) of fixation durations per subject and condition;
- an overall MFD and SD per subject, aggregating across all conditions;
- aggregated MFD per emotion across all observers.

In this project we focus on the following main features:

- (1) **Total number of fixations** per subject and emotion condition, for both Settings A and B.
- (2) **Mean Fixation Duration (MFD)** per subject and emotion — our primary fixation-based metric.
- (3) **Standard deviation of fixation durations** per subject and emotion, capturing variability in fixation timing.
- (4) **Aggregated MFD per emotion** across all observers, with corresponding SD, summarizing emotion-level differences.
- (5) **Overall MFD per subject** across all emotions, with SD, providing a subject-level summary of processing style.

These values are provided in [this feature folder link](#) which contains the files `Group_I_features_setting_A_per_observer.csv` and `Group_I_features_setting_B_per_observer.csv`. Tables 1 and 2 present snippets of the Setting A results, showing MFDs (in ms) and their corresponding SDs for each emotion and subject.

Table 1: Snippet of the Setting A csv file. Values are mean fixation durations (MFD; ms) per emotion and overall, for each subject.

subject_id	Happy	Sad	Fear	Neutral	Angry	Overall
Kashif	766.9	736.8	847.6	802.8	816.6	792.3
Kidu	525.1	422.4	463.6	376.8	534.3	458.2
Ryan	528.0	445.1	478.1	679.3	455.7	505.2
Khang	620.1	321.6	416.4	403.1	572.0	444.8
Zeeshan	457.5	393.2	408.6	409.7	389.7	410.9

Inspection of Table 1 reveals pronounced individual differences in fixation behavior. Kashif exhibits consistently longer MFDs than the other observers (Overall MFD ≈ 792 ms), suggesting a more sustained inspection strategy. In contrast, Zeeshan and Khang show

Table 2: Standard deviation (SD) of fixation durations (ms) per emotion and overall, for each subject (Setting A).

subject_id	Happy	Sad	Fear	Neutral	Angry	Overall
Kashif	725.6	734.2	893.2	959.6	707.8	801.5
Kidu	601.2	692.1	521.7	421.1	728.5	596.3
Ryan	400.4	439.1	586.0	968.6	359.4	562.7
Khang	544.5	253.5	337.6	472.8	526.2	433.5
Zeeshan	431.5	378.0	351.1	374.8	313.7	369.7

shorter overall MFDs (approximately 411 ms and 445 ms), reflecting faster, more fragmented sampling.

Across emotions, MFDs for most observers vary within a broad range of roughly 300–850 ms. Fear and neutral expressions frequently elicit relatively long fixations, whereas sad expressions tend to yield shorter ones, indicating that fixation timing is sensitive to the displayed emotion.

Table 2 complements these observations by quantifying variability in fixation duration. Kashif’s fixation durations not only have the highest means but also the largest overall SD (≈ 802 ms), reflecting substantial trial-to-trial variance. Kidu and Ryan also show relatively high variability, while Khang and Zeeshan exhibit lower SDs (≈ 434 ms and 370 ms), indicating more homogeneous fixation behavior.

Together, the MFD and SD summaries capture both emotion-level effects and robust inter-individual differences in oculomotor strategies during emotion recognition.

4.2 Visualization of Gaze Behavior

To interpret the quantitative measures we created several visualizations:

- **Scanpaths** showing sequences of fixations and saccades for a representative observer and trial from each emotion;
- **Fixation overlays** plotting all fixations for a given emotion across observers, with point size or color indicating fixation duration;
- **Gaze heatmaps** displaying the spatial density of gaze samples per emotion;
- **MFD bar charts** showing MFD by observer and emotion and aggregated MFD by emotion;
- **I-DT comparison plots** summarizing differences between Settings A and B in terms of fixation counts and MFD.

These plots allow us to connect the numerical summaries with intuitive spatial patterns in gaze behavior.

5 Results

This section presents the qualitative and quantitative outcomes of the eye-tracking analysis. For each visualization class, we first show *single-trial results for one observer (Kashif)* across all *15 stimuli (5 emotions $\times$ 3 images)*. We then provide *combined visualizations* aggregated across all five observers where appropriate, highlighting group-level patterns. This two-level structure allows us to examine both individualized gaze behavior and generalizable trends across participants.

5.1 Scanpaths

For each of the 15 facial images shown to the observers, we generated a detailed scanpath representation. Figure 2 presents all scanpaths for **Kashif** across the 15 stimuli (5 emotions $\times$ 3 identities). Each plot visualizes the temporal sequence of fixations and saccades: circles denote fixation locations (with radius proportional to fixation duration), and arrows indicate saccadic transitions.

Across all images, scanpaths exhibit a highly consistent spatial organization. Regardless of identity or emotion, Kashif’s gaze is rapidly drawn toward the *eyes* and *mouth*, after which saccades alternate between these two regions. This behavior aligns with well-established findings in face perception research, where these features carry the strongest diagnostic cues for emotion recognition. Notably, scanpaths remain concentrated within a narrow central region of the face, with minimal exploration of the forehead, cheeks, or background.

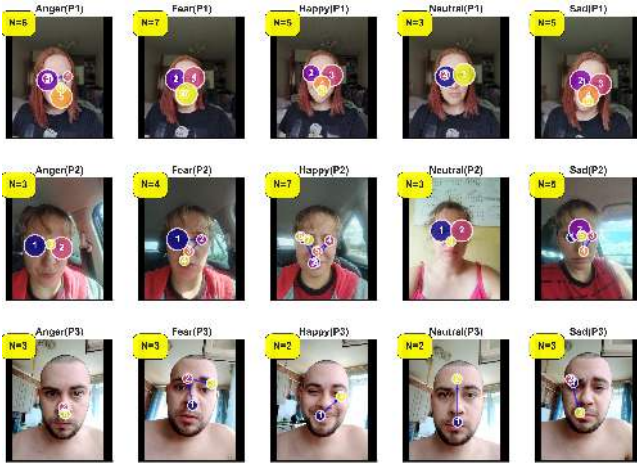


Figure 2: Scanpaths for observer Kashif across all 15 stimuli. Larger circles indicate longer fixations; arrows denote saccades. Across identities and emotions, gaze consistently cycles between the left eye, right eye, and mouth, demonstrating a stable and targeted inspection strategy.

To complement the full-set visualization, Figure 3 shows one representative scanpath per emotion for the same observer. These compact scanpaths further illustrate that the majority of fixations begin near the eyes and terminate around the mouth, with very limited deviation outside the central facial triangle.

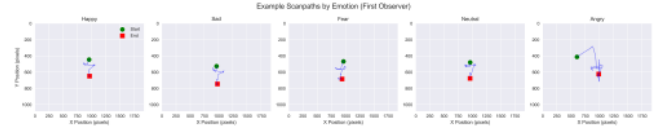


Figure 3: Representative scanpaths for Kashif across five emotional expressions. Green and red markers denote the first and final fixations. In all cases, gaze remains confined to the eye–nose–mouth region, indicating focused extraction of emotionally relevant information.

Taken together, the scanpath patterns demonstrate that gaze is strongly anchored to emotionally diagnostic features. Across all emotions, scanpaths remain localized around the eyes and mouth, displaying a highly structured and efficient viewing strategy. This qualitative behavior supports our hypothesis that observers rely primarily on these regions when identifying facial expressions.

5.2 Fixations

5.2.1 Single-Trial Fixations.

Fixations were identified using the I-DT algorithm. Across the 15 stimuli viewed by observer Kashif, fixations consistently cluster within a narrow central band containing the eyes, nose bridge, and mouth. Figure 4 shows the fixation distributions across all stimuli for this observer.

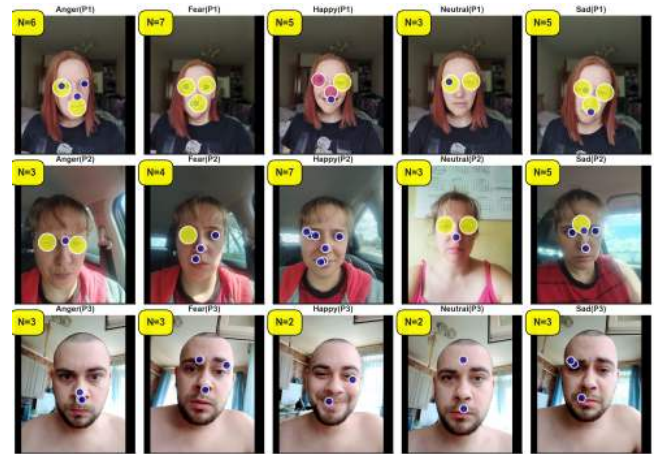


Figure 4: Fixation distribution for observer Kashif across all 15 stimuli. Each circle represents a fixation, with color and size proportional to fixation duration. Fixations concentrate strongly around the eyes and mouth, reflecting targeted processing of the most diagnostic emotional features.

Overall, the longest fixations consistently occur on the eyes, aligning with their primary role in emotional inference. The mouth also attracts substantial fixation time, particularly for *Happy* and *Angry* expressions where lip curvature or tension conveys key emotional cues. Peripheral facial areas receive markedly fewer fixations, indicating efficient and feature-focused viewing behavior.

5.2.2 Fixation Overlays by Emotion (All Observers).

To examine cross-observer patterns, fixations from all five participants were pooled by emotion (Figure 5). Each point represents a fixation centroid, color-coded by fixation duration.

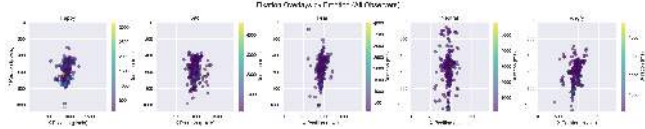


Figure 5: Fixation centroids across all observers for each emotion. Colors represent fixation durations. Observers consistently fixate the core facial region (eyes–nose–mouth). Fear and Neutral faces exhibit slightly greater vertical spread, suggesting increased exploratory behavior under uncertain or ambiguous expressions.

Across emotions, a highly robust pattern emerges: observers focus almost exclusively on the central diagnostic features of the face. However, subtle emotion-specific differences are visible. *Fear* and *Neutral* expressions show increased vertical dispersion, possibly reflecting greater perceptual ambiguity, prompting observers to sample both eyes and mouth more frequently. In contrast, *Happy* and *Angry* faces—where diagnostic cues are more salient—produce tighter fixation clusters.

5.3 Raw Gaze Samples



Figure 6: Raw gaze samples from Kashif across all 15 stimuli. Each red point corresponds to a recorded gaze sample. Although noisier than fixation-level representations, the raw traces clearly show early and repeated orienting toward the eyes, nose bridge, and mouth.

Raw gaze samples provide a fine-grained view of moment-to-moment eye movements before fixation parsing. Figure 6 displays the full gaze trajectories for one observer (Kashif) across all 15 images. Although individual samples exhibit natural noise, microsaccades, and small fluctuations in tracking accuracy, the overall spatial structure is remarkably stable across identities and emotions.

Across all images, the densest clusters of raw points consistently appear around:

- both eyes (primary concentration),
- the nose bridge,
- the upper mouth region.

Occasional samples extend toward the forehead or outer cheeks—most commonly in *Fear* and *Neutral* expressions—suggesting brief exploratory movements when emotional cues may be less immediately identifiable.

Importantly, the raw gaze distributions strongly mirror the fixation-level findings: even without algorithmic segmentation, the majority of gaze activity is concentrated within the central facial triangle. This confirms that fixation-based summaries and heatmap representations accurately reflect the underlying moment-to-moment viewing behavior, and that the observers relied on highly consistent facial regions for emotion recognition across all trials.

5.4 Heatmaps

5.4.1 Single-Trial Heatmaps.

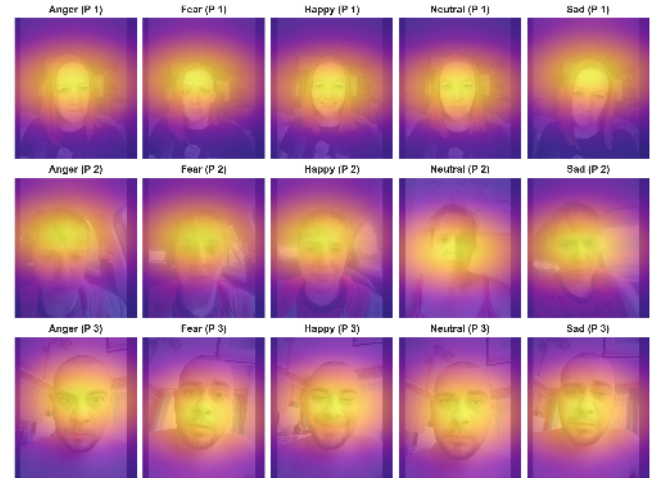


Figure 7: Gaze heatmaps for all 15 stimuli from a single observer (Kashif). Warmer colors indicate higher gaze density. Across all identities and emotions, gaze concentration forms a stable band around the eyes, nose, and upper mouth.

Heatmaps were computed by spatially aggregating gaze points to reveal dominant regions of attention. Figure 7 illustrates heatmaps for a single observer across all 15 images. Despite variations in identity and expression, the gaze distribution exhibits a strongly consistent topology:

- The **eyes** form most intense hotspot across nearly all stimuli.
- The **nose bridge** and **upper mouth** receive the next-highest density.
- Peripheral regions such as cheeks, chin, and background receive minimal gaze.

Fear and *Neutral* images display slightly more vertical spread toward the forehead and lower facial regions, suggesting broader information sampling when emotional cues are ambiguous or subtle.

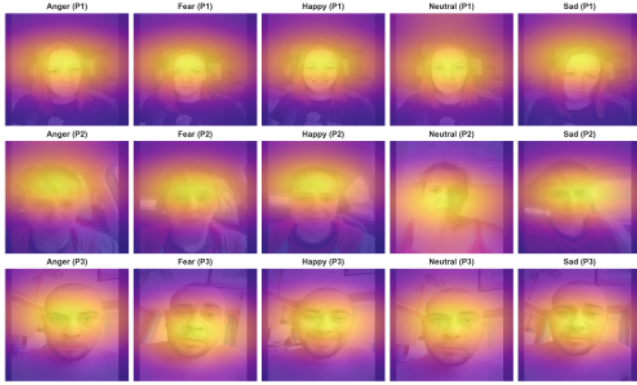


Figure 8: Combined heatmap, aggregated across all observers. The shared hotspot structure highlights that participants converged on similar facial regions when interpreting the displayed emotion.

Figure 8 shows an aggregated heatmap from multiple observers for one representative stimulus. The near-identical concentration patterns across participants suggest a shared perceptual mechanism: observers rely on a similar set of facial features—primarily the eyes and nose—when decoding emotional expressions.

5.4.2 Heatmaps by Emotion (All Observers).

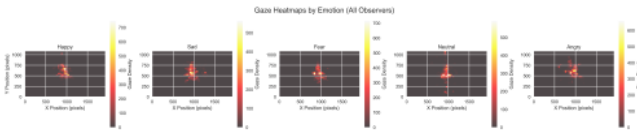


Figure 9: Heatmaps aggregated by emotion across all observers. All emotion categories show a dominant central hotspot over the eyes and nose. Happy and Angry display more compact focus, while Fear and Neutral show slightly broader vertical dispersion.

These emotion-wise heatmaps reveal subtle but interpretable differences:

- **Happy and Angry:** tighter, more compact hotspots—participants rely on highly diagnostic facial muscle cues.
- **Fear and Neutral:** broader vertical spread—likely reflecting uncertainty and additional scanning to resolve ambiguous cues.
- **Sad:** a moderately narrow band centered on the eyes and upper nose, with fewer mouth-directed fixations.

Overall, heatmap analysis reinforces the fixation and raw-gaze findings: observers prioritize the same core set of facial features, but emotional ambiguity modulates the scope of exploratory behavior.

5.5 Mean Fixation Duration by Emotion and Observer

Figure 10 (top) shows the Mean Fixation Duration (MFD) for each observer across the five emotion categories (Setting A). Clear individual differences are visible: Kashif consistently demonstrates the longest MFDs (often exceeding 800 ms), whereas Zeeshan and Kidu show shorter and more uniform fixation durations across conditions. Despite these observer-level heterogeneities, several consistent emotion-dependent patterns emerge:

- **Happy and Angry** expressions generally elicit the longest fixations.
- **Sad** expressions consistently produce the shortest fixations for most observers.
- **Fear and Neutral** fall between these extremes, but vary more strongly across individuals.

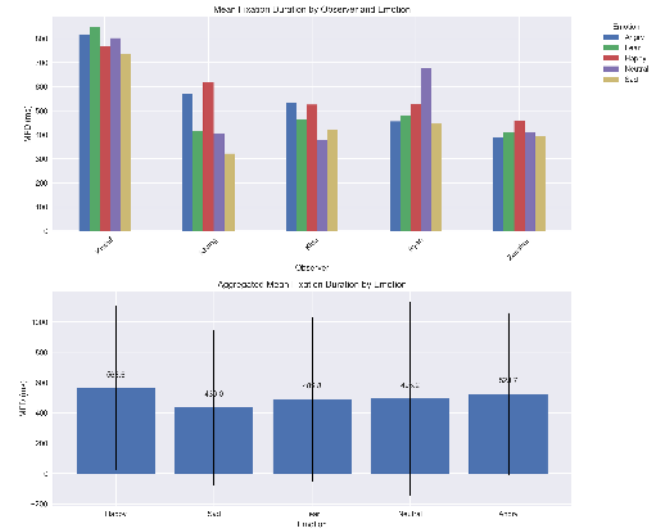


Figure 10: Top: Mean Fixation Duration (MFD) by observer and emotion (Setting A). Bottom: Aggregated MFD by emotion across all observers, with error bars showing between-subject variability. The consistency of emotion-specific differences suggests stable perceptual strategies across the group.

The bottom panel of Figure 10 summarizes MFDs aggregated across all observers. A clear ranking emerges:

- **Happy** faces produce the longest average MFD (≈ 563 ms),
- followed by **Angry** (≈ 529 ms),
- **Fear** and **Neutral** show intermediate durations (≈ 490 ms),
- and **Sad** faces elicit the shortest average fixations (≈ 436 ms).

This pattern suggests that emotional expressions differ in the amount of visual evidence observers extract before moving their gaze. Expressions such as *happy* and *angry* contain strong, salient cues—especially in the eyes and mouth—leading to longer, more confirmatory fixations. Conversely, *sad* expressions evoke shorter fixations, possibly because their visual features are less dynamic or less visually complex, reducing the need for prolonged inspection.

Taken together, the MFD results highlight emotion-dependent differences in perceptual processing and reinforce the fixation and

heatmap findings: observers engage most deeply with facial expressions that contain strong diagnostic cues.

5.6 Comparison of I-DT Settings

Figure 11 summarizes the differences between Setting A and Setting B. The upper panels compare mean fixation duration (MFD) and fixation counts across emotions, while the lower panel shows overall MFD differences across observers.

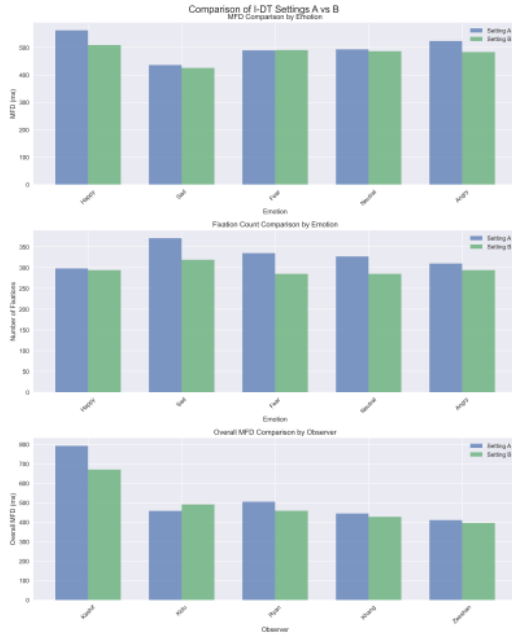


Figure 11: Comparison between I-DT Settings A and B. Top-left: MFD by emotion; top-right: fixation count by emotion; bottom: overall MFD by observer. Setting B (stricter dispersion and duration thresholds) produces fewer fixations and slightly lower MFD estimates, demonstrating the sensitivity of fixation metrics to algorithm parameters.

Summary of Differences by Emotion. Table 3 summarizes how fixation count and MFD change when switching from Setting A to Setting B. Across all emotions, the stricter Setting B consistently detects fewer fixations and yields lower MFDs. Reductions are largest for *Sad* and *Fear*, suggesting that these expressions elicit more short or spatially dispersed fixations—exactly the type most likely to be removed when thresholds become stricter.

Emotion	Fixations (B vs. A)	MFD (B vs. A)
Happy	Decrease	Moderate decrease
Sad	Largest decrease	Moderate decrease
Fear	Large decrease	Slight decrease
Neutral	Moderate decrease	Small decrease
Angry	Moderate decrease	Moderate decrease

Table 3: Qualitative comparison of fixation count and MFD changes from Setting A to Setting B across emotions.

Summary of Differences by Observer. Observers varied substantially in how sensitive their fixation profiles were to the parameter shift (Table 4). *Kashif* shows the largest reduction in overall MFD, indicating that a substantial proportion of his fixations fall close to the threshold boundaries. Conversely, *Zeeshan* exhibits only minimal changes, suggesting more stable or longer fixations that remain above threshold in both settings.

Observer	Fixations (B vs. A)	Overall MFD (B vs. A)
Kashif	Decrease	Large decrease
Kidu	Slight decrease	Small decrease
Ryan	Moderate decrease	Moderate decrease
Luel	Moderate decrease	Moderate decrease
Khang	Slight decrease	Small decrease
Zeeshan	Slight decrease	Very small decrease

Table 4: Observer-level summary of changes from Setting A to Setting B.

Interpretation. These results highlight that fixation detection is highly sensitive to algorithmic parameterization. Tighter spatial and temporal thresholds (Setting B) systematically reduce the number of detected fixations and lower overall MFD estimates. Crucially, the magnitude of these effects varies across both emotions and observers, reflecting:

- differences in the spatial distribution and stability of gaze for each expression, and
- individual differences in oculomotor behavior (e.g., tendency to make shorter, more frequent fixations).

The comparison demonstrates the importance of explicitly reporting fixation detection parameters and, whenever possible, validating robustness by analyzing multiple parameter configurations.

6 Discussion

Across all analyses, a consistent picture of facial information processing emerges. Both the single-trial visualizations and the aggregated overlays show that observers concentrate their gaze within a narrow, highly diagnostic region of the face—primarily the left eye, right eye, nose bridge, and upper mouth. This spatial pattern appears across all emotions and identities, and remains visible even in the raw gaze traces, demonstrating that the overall gaze strategy is highly stable and does not depend strongly on fixation segmentation.

The temporal measures reinforce this interpretation. Mean Fixation Duration (MFD) varies systematically across emotions, with happy and angry expressions eliciting longer fixations than sad expressions. These longer dwell times may reflect increased emotional salience, deeper engagement, or the need to extract more detailed information from these expressions. The intermediate MFD values for fear and neutral expressions align with the more vertically diffuse fixation patterns seen in the overlays and heatmaps, suggesting a mixture of targeted inspection and exploratory sampling.

Importantly, the comparison of I-DT parameter settings reveals that fixation metrics are shaped not only by observers’ gaze behavior but also by the algorithmic thresholds used to detect fixations. Stricter settings reduce both the number of detected fixations and overall MFD, with the magnitude of reduction varying by emotion

and by observer. This sensitivity underscores a key methodological point: fixation statistics cannot be interpreted independently of their detection parameters. Even in a relatively simple student project, differences in dispersion and duration thresholds meaningfully alter conclusions about gaze stability and processing time.

Taken together, the study reproduces several well-established findings in the eye-tracking literature on face perception [1, 2]: observers tend to adopt a stable, feature-focused viewing strategy, and fixation timing reflects both stimulus characteristics and the cognitive demands of emotion interpretation. The results demonstrate how a carefully designed analysis pipeline—incorporating raw gaze checks, fixation detection, spatial visualizations, and sensitivity analyses—can yield a coherent, scientifically grounded understanding of gaze behavior.

7 Limitations and Future Work

Several limitations should be considered when interpreting the findings. First, the participant sample is small (five observers), which limits generalizability and reduces statistical power for detecting subtle differences between emotional categories. Second, all stimuli consisted of static photographs. Dynamic expressions or more naturalistic videos may evoke richer or more varied gaze patterns, especially for emotions such as fear or anger that involve movement cues. Third, the present analysis relied on fixation centroids without explicitly defined Areas of Interest (AOIs). Although the spatial patterns were clear, an AOI-based analysis could quantify more precisely how often and how long observers fixate specific facial regions such as the left eye, right eye, nose, or mouth.

Methodologically, we examined only the dispersion-threshold (I-DT) algorithm. While the comparison between two parameter settings demonstrated the importance of algorithmic choices, future work could incorporate alternative fixation detection approaches, such as velocity-based methods (e.g., I-VT) or machine-learning-based segmentation, to assess cross-algorithm consistency.

Future extensions could therefore include (i) a larger and more diverse participant pool, (ii) a broader stimulus set with dynamic expressions, (iii) AOI-based metrics to complement spatial overlays, (iv) comparisons across multiple fixation detection algorithms, and (v) integration of behavioral measures such as recognition accuracy or response time. Together, these enhancements would strengthen the ecological validity of the study and deepen our understanding of how visual attention supports emotion recognition.

Group Members and Contributions

Table 5 summarizes the project stages and their responsibilities. Table 6 reports the contribution percentages for each group member. Homework contributions are normalized to 100% across all members, and project contributions are normalized to 100% across the final project work, as required by the assignment.

Table 5: Project stages and main responsibilities.

Stage	Responsibility description
1	Experiment design and hypothesis formulation
2	Experiment setup and coding with the eye tracker
3	Data collection (running participants)
4	Data analysis coding (cleaning, I-DT A/B, features, plots)
5	Report writing, editing, and final checking

Table 6: Group member contributions for homeworks and final project, with summary of responsibilities by stage index.

Member	Homeworks (%)	Project (%)	Summary (Stages)
Anirban Das	12.5	8.00	1, 5
Kashif Ali Javaid	12.5	20.00	1, 3, 4, 5
Muhammad Zeeshan	12.5	12.00	1, 3, 5
Luel Gebre	12.5	12.00	1, 3, 5
Kidu Welegerima	12.5	20.00	1, 2, 3, 5
Minh Khang Tran	12.5	20.00	1, 3, 4, 5
Abdullah Al Ryan	12.5	12.00	1, 3, 5
Farooq Kadri	12.5	8.00	1, 5

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